LifeBand: A Long-Range, Low-Power Wearable for Real-Time Aquatic Distress Detection

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Keywords: LoRa 915 MHz · ESP32 Edge Sensor Fusion · PPG / IMU / GPS · Search-and-Rescue Telemetry

July 4, 2025: Camp Mystic and the Guadalupe River Flood

26 ft

Guadalupe River rise in about 45 minutes before dawn

27

Camp Mystic campers and counselors lost (25 campers, 2 counselors)

138

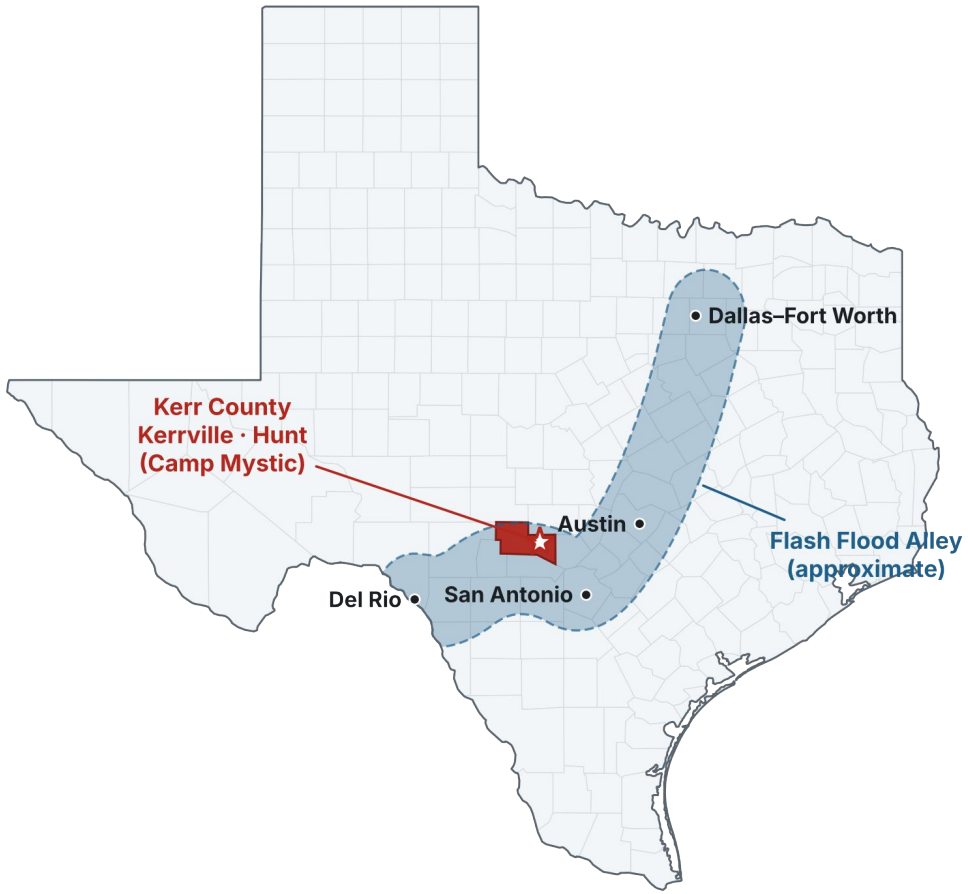
lives lost along the Guadalupe River, per Texas reporting (Sept. 2025)

TIMELINE

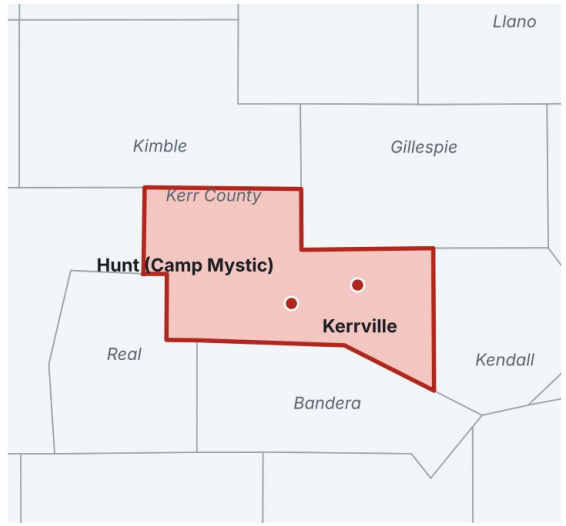
JUL 4, 2025	Flash-flood emergency in Kerr County. The Guadalupe rises 26 ft in about 45 minutes, sweeping through homes, RV parks and summer camps, including Camp Mystic near Hunt.
JUL 5–9, 2025	Search and rescue continues as the toll passes 100. Kerr County officials say the county had no flood warning system; a federal major disaster declaration is signed.
AUG 2025	Reporting finds Camp Mystic is one of 13 camps on the Guadalupe with structures on flood-prone land; the two cabins where most victims slept sat in the river floodway.
SEP 2025	Texas moves camp-safety reforms: overnight cabins out of floodplains, radios and alert systems for weather warnings, and staff emergency training.
JUN 18, 2026	State investigators' final report: no state-compliant written emergency plans, counselors' phones confiscated without backup radios, and hours of storm alerts not acted on.
JUL 2026	One year later, new flash-flood emergencies hit Kerrville, Hunt and Uvalde; the Guadalupe at Hunt rose 10 ft in a single hour.

Sources: CBS News (Jul 2025); KWTX complete timeline (updated Jul 2026); Texas Tribune and KSAT on the state legislative report (Jun 2026); ABC News (Jul 2026).

Where It Happened: Kerr County in Flash Flood Alley



Texas with Kerr County (red) and the approximate extent of Flash Flood Alley (blue).



KERR COUNTY

Kerrville is about 70 miles northwest of San Antonio, in the Texas Hill Country.

Camp Mystic sits on the Guadalupe River at Hunt, west of Kerrville.

On July 4, 2025 the river rose about 26 ft in 45 minutes.

Flash Flood Alley	A crescent from near Dallas through Austin and San Antonio, curving west toward Del Rio (I-35 corridor, Balcones Escarpment).
Why it floods	Steep terrain, shallow semi-arid soils and intense rainfall send runoff racing into fast-rising creeks and rivers.
Texas leads the nation	1,069 flood deaths from 1959 to 2019, the most of any state; 58% involved vehicles.
Local impact for LifeBand	Pools, camps and river corridors across the alley, including the Austin area, can use an infrastructure-free safety layer.

Map: U.S. Census county boundaries. The Flash Flood Alley band is illustrative, based on Scientific American and Dallas Morning News descriptions, not an official boundary. Flood statistics: Scientific American.

What Went Wrong, and the Gap LifeBand Targets

Documented failure	Source	LifeBand design response
Hours of storm alerts without effective monitoring or evacuation	Texas legislative investigators, Jun 2026	Automated detection: the wearable flags submersion, impact and abnormal vitals itself, and the dashboard raises a siren and map alert without anyone having to notice. Roadmap: inbound hazard alerts to the wearer.
Counselors' phones confiscated; no backup radios	Texas legislative investigators, Jun 2026	Independent 915 MHz LoRa radio link: no phone, cellular or Wi-Fi needed. Field-tested to 1 km, designed for 10+ km.
Chaotic accounting of who was missing and where; conflicting information	Texas legislative investigators, Jun 2026	Every packet is logged in SQLite with device ID, timestamp and GPS; a live map pins each wearer, color-coded by urgency score, as one shared source of truth.
No flood warning system in Kerr County	Judge Rob Kelly, Jul 2025 (KWTX)	Base station runs locally (Python FastAPI + SQLite): no cloud and no cellular needed, on a low-cost kit suited to camps and rural facilities.

Why it matters for Texas: 13 summer camps along the Guadalupe have structures on flood-prone land (Texas Tribune, Aug 2025), and the river flooded again in July 2026. Low-cost safety tech that works when infrastructure fails is needed now.

LifeBand is a complementary safety layer; it does not replace evacuation planning, flood sirens or trained staff.

Problem Statement

Drowning and inland flash flooding remain leading causes of accidental aquatic mortality in the U.S., with Central Texas' “Flash Flood Alley” representing a geographically concentrated high-risk corridor.

Metric	Value	Source Context
Annual U.S. drowning deaths	4,000+	National incidence baseline
TX residential/municipal pool drownings / yr	80+	Quiet submersion, delayed response
TX aquatic fatalities from non-coastal flash floods	43%	Statewide distribution
Historic TX flood fatalities (nationwide-leading)	1,000+	Cumulative recorded casualties

Table 1 — Aquatic safety incidence statistics (U.S. and Texas).

CASE REFERENCE

Camp Mystic Flash Flood — Guadalupe River, Kerr County, TX (July 2025)

Rapidly rising water swept through the camp during early morning hours, resulting in 27 fatalities among campers and staff.

Systemic vulnerability identified:

When local infrastructure fails, responders lose both communication channels and real-time situational awareness of victims — in flash floods and silent pool submersions alike.

Source: CBS News, “Death toll from Texas flash floods,” July 2025.

Limitations of Existing Emergency-Response Systems

Bottleneck	Mechanism	Operational Consequence
Signal attenuation in water	Cellular / Wi-Fi operate at high RF frequencies with poor penetration through water, foliage, and saturated atmosphere	Unreliable connectivity in open water and flash-flood conditions
Reliance on manual activation	SOS-button paradigm requires conscious, voluntary user action	Fails during submersion, impact shock, or loss of consciousness
Cost & infrastructure dependency	Satellite links and dense mesh networks require recurring subscriptions or power-heavy infrastructure	Impractical for municipal pools, rural facilities, and youth camps

Table 2 — Summary of technical bottlenecks in cellular-, Wi-Fi-, and satellite-dependent aquatic safety systems.

Sources: "Signal Attenuation," ScienceDirect Topics; "RF Path and Absorption Loss Estimation for Underwater Wireless Sensor Networks," PMC (NIH).

System Architecture Overview

LifeBand couples an edge-computing wearable node with an offline 915 MHz LoRa uplink and a search-and-rescue monitoring backend — requiring no cellular or Wi-Fi infrastructure.

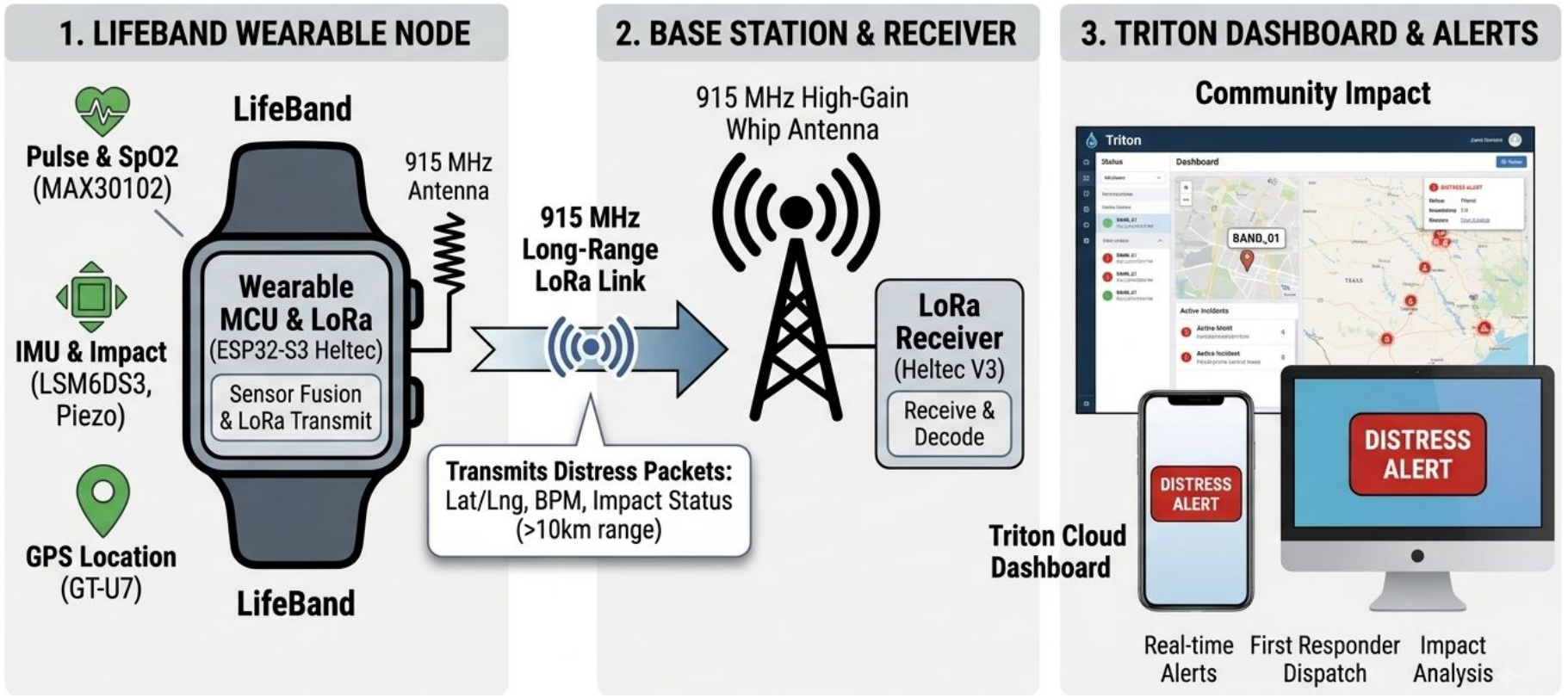


Figure 1 — End-to-end telemetry path: wearable sensor node → LoRa base station → Triton cloud dashboard (author's system diagram).

Hardware Specification

Subsystem	Component	Interface	Function
Core MCU	ESP32 Heltec V3 (LoRaWAN)	GPIO / SPI / I²C	Real-time on-device sensor fusion & threshold logic
Biometric sensor	MAX30102 (PPG / SpO_2)	I²C	Pulse rate & oxygen saturation — tachycardia / cessation detection
Impact sensor	Piezoelectric element	Analog	Hydro-barometric pressure spikes from sudden water entry
Kinematic sensor	LSM6DS3 6-axis IMU (accel/gyro)	I²C	Orientation change, free-fall, active-drowning motion signature
Positioning	GT-U7 GPS/GNSS module	UART	Lat / long / altitude embedded in distress payload
Radio	SX1262 LoRa transceiver	SPI	915 MHz sub-GHz long-range offline telemetry

Table 3 — Hardware component specification.

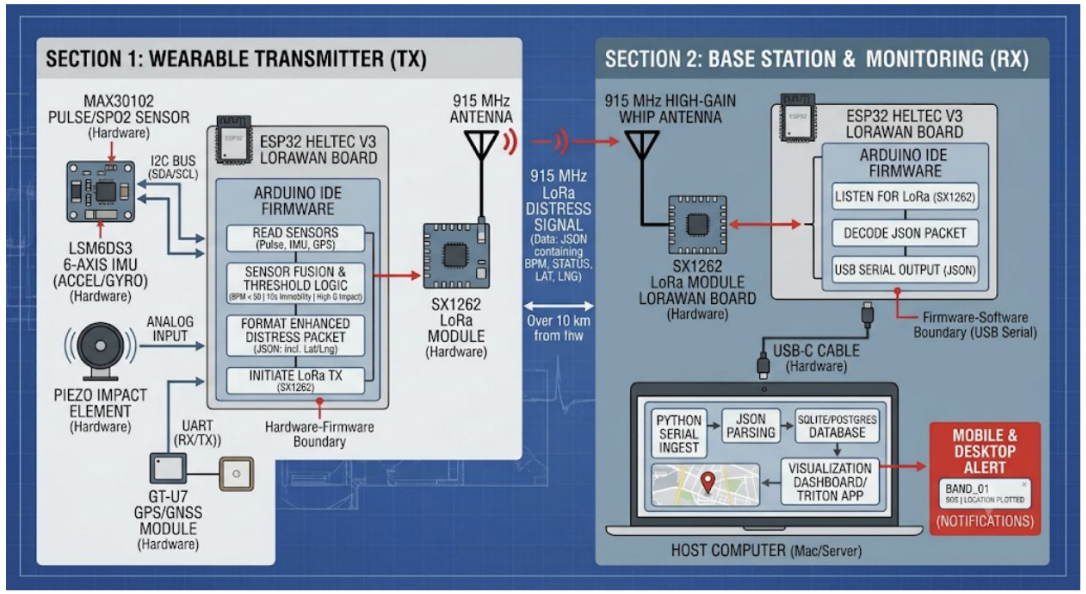


Figure 1b — Hardware Block Diagram

Firmware-hardware boundary on the wearable transmitter (left) and the base-station receiver / host-computer pipeline (right), matching the components in Table 3.

Autonomous Distress Detection

EMERGENCY TRIGGER CONDITION

$$f(t) = w_1 \cdot \Delta HR + w_2 \cdot (dP/dt) + w_3 \cdot |dz/dt| \qquad \text{trigger when } f(t) > T(\text{alert})$$

Term	Definition	Detection Basis
ΔHR	Biometric deviation	PPG pulse variance vs. resting baseline — extreme tachycardia or cardiac cessation
dP/dt	Pressure dynamics	Transient mechanical spikes from sudden hydrodynamic shock / water entry
dz/dt	Depth rate of change	Rapid vertical displacement derived from hydro-barometric pressure shift
ΔP, GPS	Kinematic & spatial offset	6-axis IMU displacement cross-referenced with absolute GPS coordinates

Table 4 — Trigger-function variable definitions.

HARDWARE OVERRIDE PATH

Manual SOS Button

A tactile switch provides an instantaneous, unconditional override: activation forces the ESP32 into an immediate alert state and broadcasts an emergency payload, bypassing f(t) evaluation entirely.

Rationale: guarantees a rescue path when automated thresholds are ambiguous — fatigue, cramping, or early-stage distress.

Backend Pipeline: Offline-First Telemetry Ingest

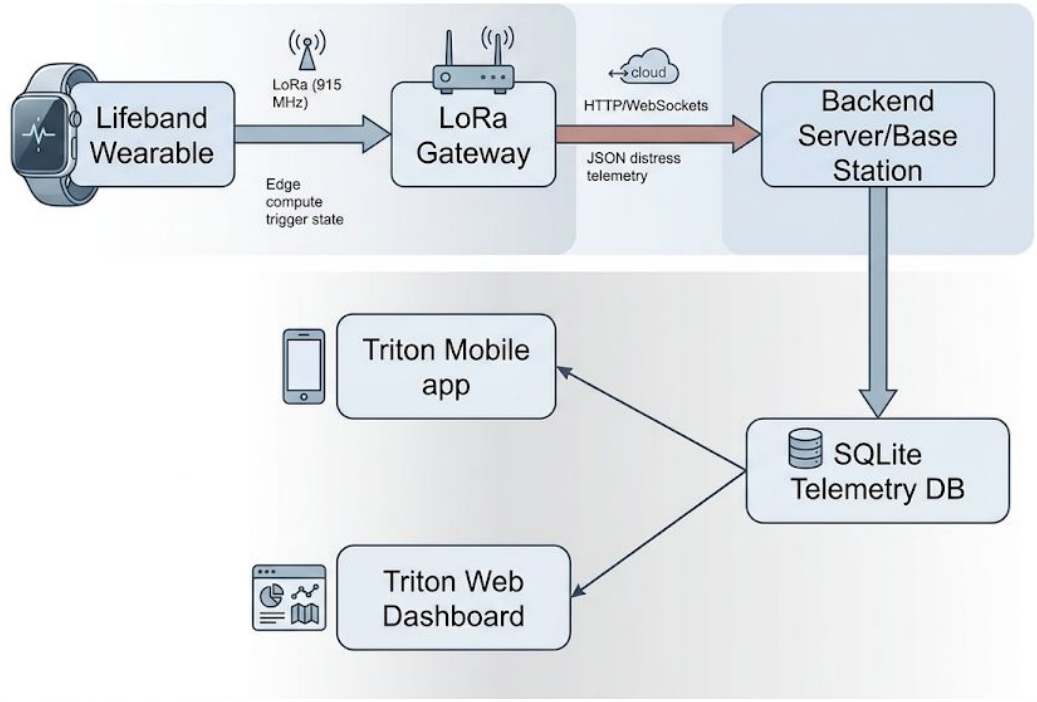


Figure 2: Telemetry data flow from wearable to LoRa gateway, FastAPI backend, SQLite and the Triton clients.

PIPELINE STAGES

01 LoRa gateway

An SX1262 on an ESP32 Heltec V3 listens for 915 MHz packets, decodes the JSON payload and writes it to USB serial.

02 Python serial ingest

Reads the serial stream and parses each JSON packet.

03 FastAPI async server

Extracts the emergency payload and pushes low-latency updates to clients over HTTP/WebSockets.

04 SQLite telemetry DB

Logs every packet instead of overwriting, so trends and post-incident audits are possible.

05 Triton mobile + web

Render alerts, map pins and vitals in real time.

Offline-first: Runs on a local base station; no cloud, cellular or Wi-Fi dependency.

Event-driven: A Distress State flag (including Manual SOS) triggers the alert cascade.

Auditable: Append-only telemetry log with device ID, timestamp and trigger source.

API Contract: One JSON Schema, Three Message Types

AUTOMATIC DISTRESS

```
{
  "device_id": "LBID-005",
  "msg_type": "DISTRESS",
  "timestamp": "2026-08-22T15:42:10Z",
  "trigger": "AUTOMATIC_SUBMERSION",
  "location": {
    "latitude": 30.448520,
    "longitude": -97.784220,
    "altitude_m": 220.1
  },
  "telemetry": {
    "heart_rate_bpm": 165,
    "spo2_percent": 88,
    "submersion_depth_cm": 110.0,
    "submersion_duration_sec": 12
  },
  "radio": {
    "spreading_factor": 8,
    "rssi_dbm": -76.8
  },
  "status": {
    "battery_percent": 88,
    "state": "CRITICAL_DROWNING"
  }
}
```

MANUAL SOS

```
{
  "device_id": "LBID-017",
  "msg_type": "DISTRESS",
  "timestamp": "2026-08-22T15:42:25Z",
  "trigger": "MANUAL_SOS",
  "location": {
    "latitude": 30.448528,
    "longitude": -97.784228,
    "altitude_m": 220.3
  },
  "telemetry": {
    "heart_rate_bpm": 178,
    "spo2_percent": 95,
    "submersion_depth_cm": 15.0,
    "submersion_duration_sec": 3
  },
  "radio": {
    "spreading_factor": 8,
    "rssi_dbm": -78.2
  },
  "status": {
    "battery_percent": 91,
    "state": "CRITICAL_DROWNING"
  }
}
```

HEARTBEAT (HEALTH-CHECK)

```
{
  "device_id": "LBID-001",
  "msg_type": "HEARTBEAT",
  "timestamp": "2026-08-22T15:40:00Z",
  "status": {
    "battery_percent": 95,
    "state": "NORMAL"
  },
  "radio": {
    "spreading_factor": 8,
    "rssi_dbm": -71.2
  }
}
```

Distress payloads carry full location and telemetry; heartbeats carry only status and radio fields. The same schema feeds the ingest server, the SQLite log and both Triton clients.

Figure 3: Mock payloads transmitted from two LifeBands to the base station (paper section 4.3).

FIELD GROUPS

`device_id` · `msg_type` · `timestamp`

Which wearer, which message type (DISTRESS or HEARTBEAT), and when.

`trigger`

AUTOMATIC_SUBMERSION or MANUAL_SOS; selects the alert path.

`location`

Latitude, longitude, altitude: pins the map.

`telemetry`

Heart rate, SpO_2, depth and submersion time feed the vitals cards and urgency score.

`radio`

Spreading factor and RSSI for link-quality diagnostics.

`status`

Battery percent and device state (e.g. CRITICAL_DROWNING, NORMAL).

Data Layer and Multi-Modal Alert Engine

SQLITE TELEMETRY SCHEMA

Column	Purpose
`device_id`	Identifies the wearer
`timestamp`	Orders events and measures elapsed time
`distress_score`	Ranks urgency and color-codes map pins
`gps_coordinates`	Locates the victim
`pulse_rate`	Feeds the heart-rate trend card
`trigger_source`	Automatic detection vs. manual SOS
`battery_life`	Drives low-battery warnings

Design choice: constant logging rather than overwriting the latest reading enables post-incident audit and trend analysis.

ALERT CASCADE ON A DISTRESS-STATE PACKET (INCL. MANUAL SOS)

Auditory + haptic

Unmuted high-frequency siren at the mobile/dashboard station for crowded pools or severe weather, with haptic feedback on clients.

Visual mapping

Interactive map pins the exact GPS position, auto-centers on the distress location and color-codes pins by distress score.

Physiological tracking

Live cards show heart-rate trend, depth displacement rate, motion state, age and device battery.

LIFE OF A DISTRESS PACKET

Wearable flags distress › **LoRa gateway decodes JSON** › **FastAPI ingest** › **SQLite log** › **Alert cascade + dispatch**

Experimental Protocol

Phase	Objective	Procedure	Metrics Collected
1. Sensor Calibration	Establish baseline threshold accuracy; eliminate false positives	Breadboard simulation — cardiac-rate changes, taps, rapid drops into water bowl	Sensor latency, false-positive rate, motion/depth-triggered spikes
2. RF Transmission	Quantify LoRa signal strength & delivery across obstruction types	RSSI / packet-loss logged 10 m–1 km: open air, open water, tree canopy, urban	RSSI, Packet Delivery Ratio, packet delivery time (latency)
3. Power Model	Quantify battery longevity; compare edge vs. cloud computing	Datasheet-derived energy modeling across sleep / monitor / TX states	Average current draw, estimated battery life
4A. Incident Sim — Managed Pool	Validate edge→dashboard→mobile alerting on manual SOS	Simulated swimmer triggers tactile SOS during surface fatigue in a crowded pool	Alert latency, map-pin accuracy, multi-modal siren verification
4B. Incident Sim — Flash Flood	Validate offline LoRa range & GPS tracking during evacuation	50 simulated victims swept downstream; distress broadcast to SAR gateway	GPS accuracy, vector-update frequency, vitals/age urgency bucketing

Table 6 — Experimental protocol phases, procedures, and metrics.

RF Signal Propagation & Reliability

Distance	Environment	SF	RSSI (dBm)	PLR (%)	Latency (ms)
50 m	Open Air / Open Water / Tree Canopy / Urban	SF7	–69.0 / –73.4 / –79.5 / –84.6	0.0 / 0.0 / 0.5 / 1.2	118 / 122 / 125 / 130
200 m	Open Air / Open Water / Tree Canopy / Urban	SF7 / SF7 / SF8 / SF8	–81.0 / –86.6 / –94.5 / –101.4	0.5 / 1.2 / 4.0 / 6.8	120 / 128 / 240 / 252
500 m	Open Air / Open Water / Tree Canopy / Urban	SF7 / SF8 / SF9 / SF9	–89.0 / –95.4 / –104.5 / –112.6	1.8 / 4.4 / 9.0 / 12.7	124 / 248 / 485 / 510
1000 m	Open Air / Open Water / Tree Canopy / Urban	SF8 / SF10 / SF11 / SF12	–95.0 / –102.0 / –112.0 / –121.0	6.6 / 12.0 / 20.5 / 27.9	242 / 960 / 1,840 / 3,620

Table 7 — RF propagation measurements across distance and environment.

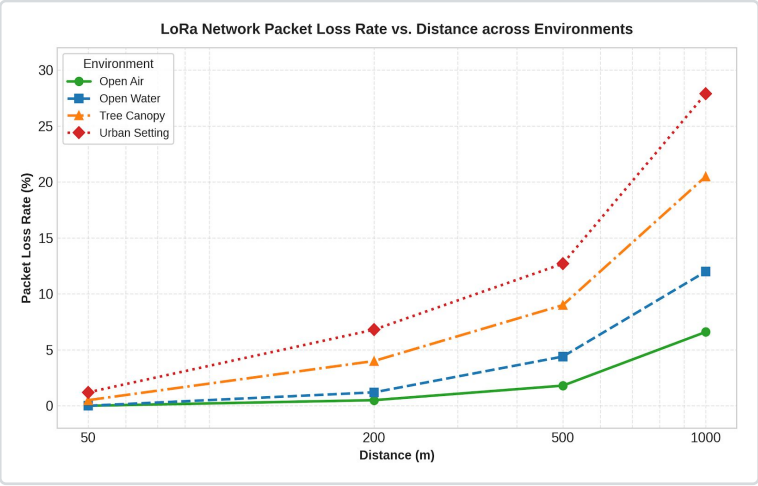


Figure 4a — Packet loss rate vs. distance.

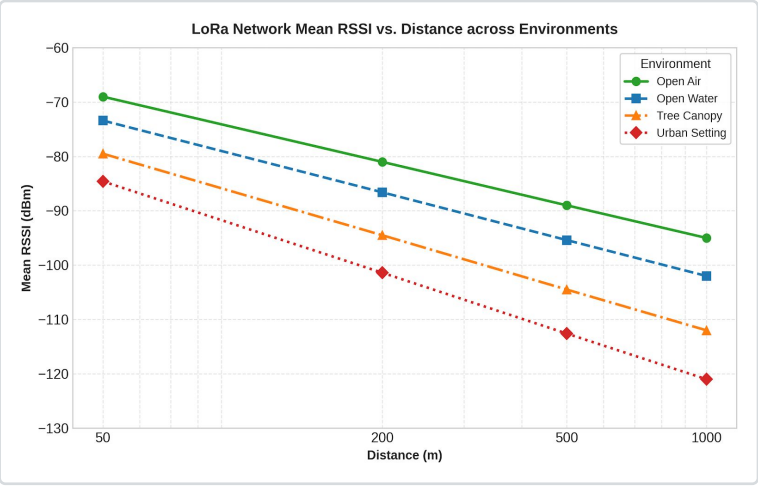


Figure 4b — Mean RSSI vs. distance.

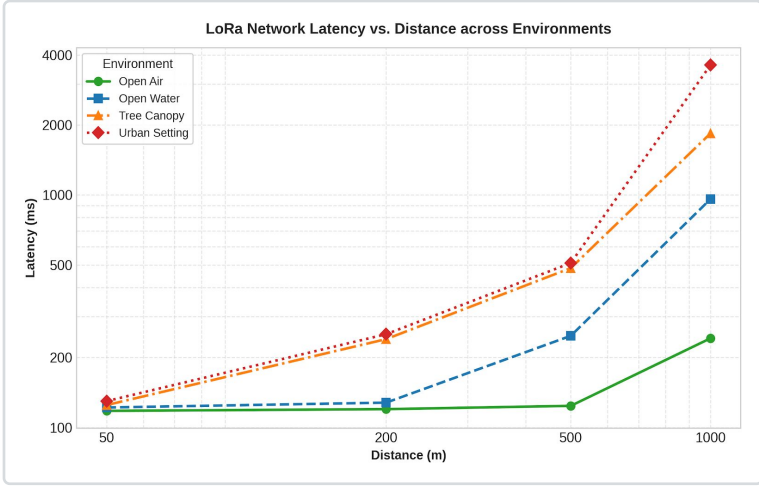


Figure 4c — Latency vs. distance.

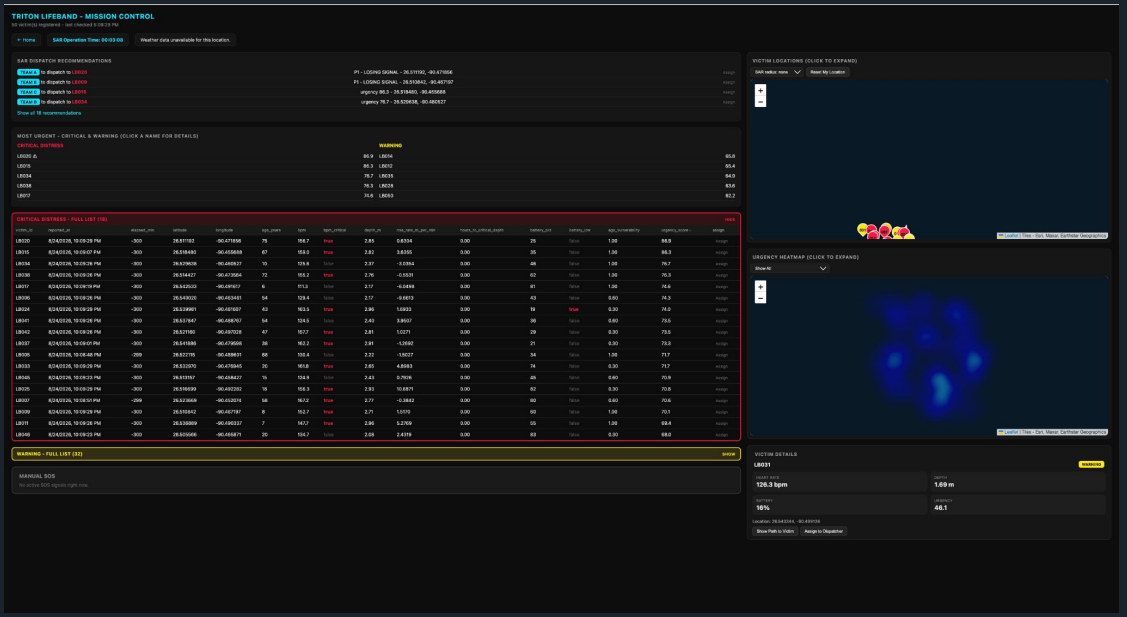
Power Model & Battery Longevity (1,000 mAh cell)

Operational Mode	Duty Cycle	Total System Current	Estimated Battery Life
Deep Sleep	100% sleep state	≈16.5–20 µA	≈4.5–5.5 years
Standby	99.9% idle state	≈25–50 µA	≈1.8–3.6 years
Normal Monitoring	Continuous passive sampling	≈1.5–3.2 mA	≈10.4–22.2 days
High-Frequency TX	Periodic burst every 30 s	≈12.5 mA (avg)	≈64 hours (≈2.6 days)
Emergency Distress — Flood	100% continuous dispatch, SF10–12	≈85–110 mA (avg)	≈7.2–9.4 hours
Emergency Distress — Pool	Rapid burst every 2–5 s, SF8	≈55–68 mA (avg)	≈11.8–14.5 hours

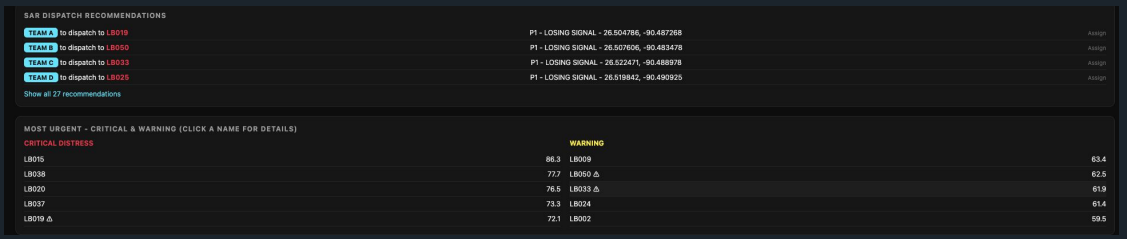
Table 8 — Power model and estimated battery life by operational mode.

Finding: shifting from SF12 (long-range flood rescue) to SF8 for localized pool environments (≤300 m) reduces packet time-on-air by >80% and extends emergency battery life from 7.2 to 11.8+ hours. On-device sensor fusion reduces total energy consumption by over 60% relative to a continuous cloud-streaming architecture.

Web Mission Control: Live Map and SAR Dispatch Queue



Map, urgency heatmap and victim details (50-node simulation).



SAR dispatch recommendations and most-urgent critical/warning list.

WHAT THIS VIEW SHOWS

Live victim map

Pins every registered wearer by GPS and refreshes as packets arrive, so responders see where people are and which way they are moving downstream.

Urgency heatmap

A density overlay of urgency scores shows where the most critical victims are clustered.

SAR dispatch recommendations

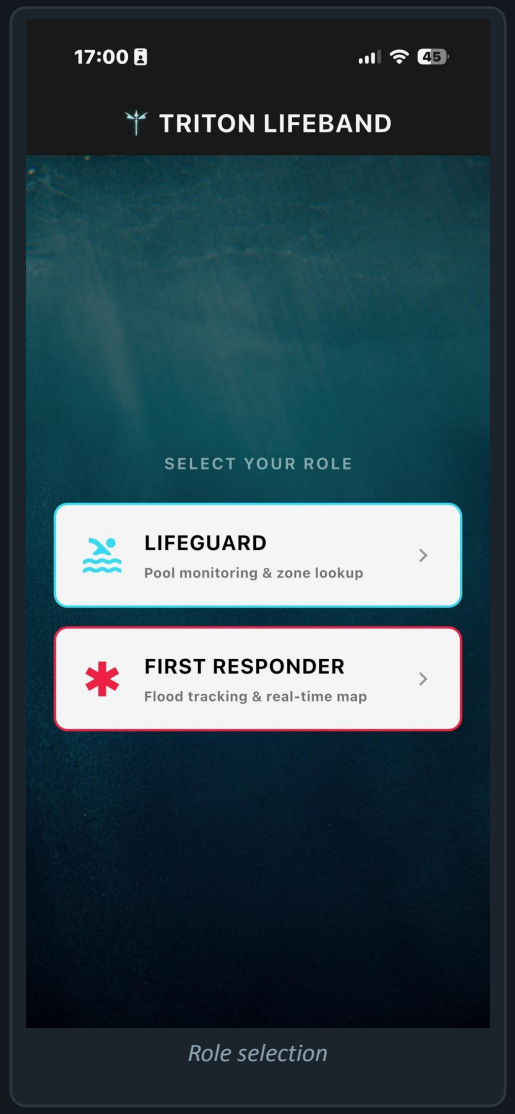
Ranks victims by urgency and suggests which team to send to which device, for example “Team A to dispatch to LB019: losing signal”.

Critical, warning and SOS lists

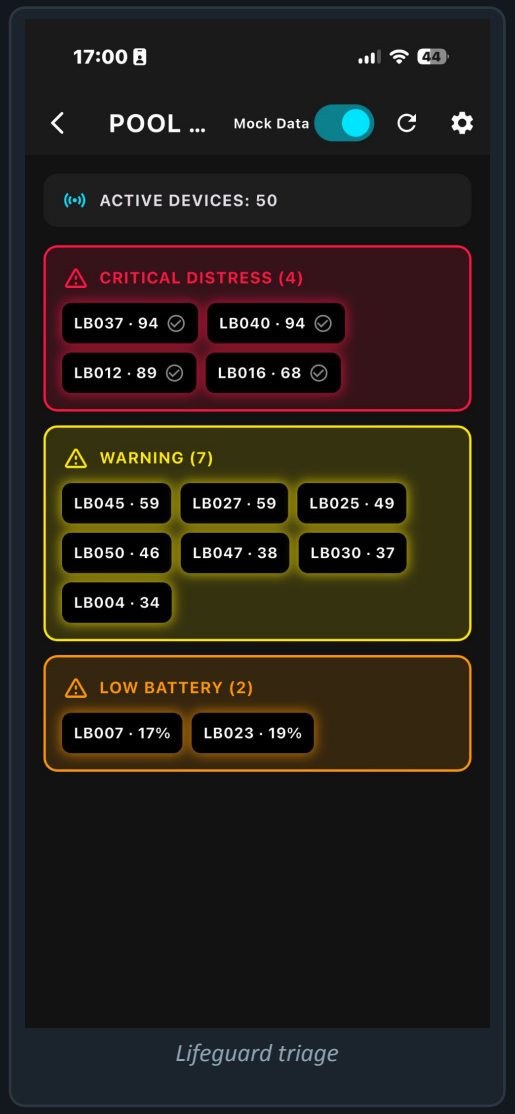
Sortable tables expose BPM, depth, rise rate, battery, age vulnerability and urgency score; active manual SOS signals get their own banner.

Simulation B: 50 victims, mock telemetry, off-grid flash-flood scenario.

Mobile App: Role Entry and Lifeguard Triage



Role selection



Lifeguard triage

WHAT THESE SCREENS SHOW

Role-based entry

The same app serves two users: Lifeguard (pool monitoring and zone lookup) and First Responder (flood tracking and real-time map).

Active-device header

Shows 50 active wearables, with a Mock Data toggle for demos and testing plus refresh and settings controls.

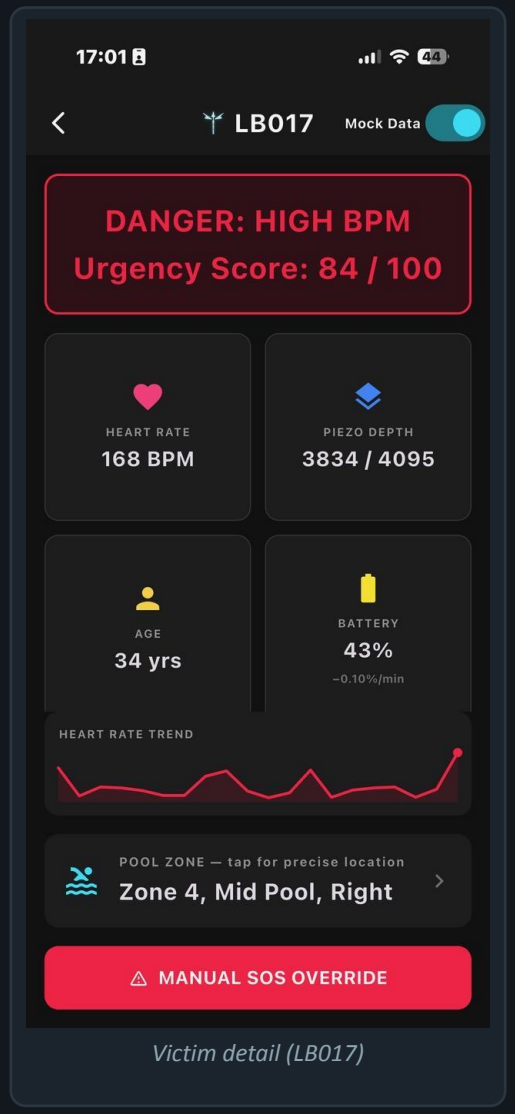
Triage tiers

Wearers are grouped into Critical Distress (4), Warning (7) and Low Battery (2). Each chip shows the device ID and urgency score, e.g. LB037 · 94.

Glanceable color coding

Red, yellow and orange tiers let a lifeguard see who needs attention first without reading numbers; selecting a chip opens that wearer's detail screen.

Mobile App: Victim Detail and Manual SOS Override



Victim detail (LB017)

WHAT THIS SCREEN SHOWS

Danger banner and urgency score

A red banner states the trigger (here “HIGH BPM”) with an urgency score of 84 / 100. The web dashboard exposes the inputs: BPM, depth, rise rate, battery and age vulnerability.

Live vitals cards

Heart rate 168 BPM, piezo depth reading 3834 / 4095, age 34 and battery 43% with its drain rate.

Heart-rate trend

A sparkline shows the spike leading up to the alert, giving context beyond a single reading.

Pool-zone lookup

“Zone 4, Mid Pool, Right” converts GPS into a pool zone; tapping gives the precise location so a lifeguard knows where to go.

Manual SOS override

A full-width control at the bottom of the screen for one-tap action on this wearer.

Summary of Engineering Contributions

Contribution	Mechanism	Result
Automated multi-sensor edge fusion	On-device threshold evaluation of PPG, IMU, piezo & GPS — no manual trigger required	>60% lower energy consumption vs. cloud-streaming architecture
Sub-GHz long-range communication	915 MHz LoRa with dynamic spreading-factor adaptation	Reliable offline links over water and through dense obstruction to ~1 km
Multi-modal alerting software	FastAPI + SQLite backend drives audio/haptic/visual alert cascade with GIS mapping	Real-time SAR dispatch recommendations from raw telemetry

Table 9 — Summary of engineering contributions.

LifeBand demonstrates a scalable, low-cost safety layer applicable to swimming-pool facilities and off-grid flash-flood search-and-rescue operations alike.

Future Works and Applications

Direction	Scope
Custom PCB integration	Multi-layer board, LiPo battery, IP68-rated waterproof silicone housing
Dynamic LoRa mesh telemetry	Wearables act as repeater nodes to extend coverage across floodplains and open water
Edge machine learning	Replace rigid threshold model with lightweight on-device neural networks trained on field data
Bi-directional safety wearable	Inbound hazard broadcasts — severe weather, dangerous currents, swim-zone boundaries
Field validation with first responders	Multi-node stress testing with emergency agencies, camps, and municipal aquatic facilities

Table 10 — Future development roadmap.

Setting	Rationale
Municipal & community pools	Continuous automated layer for lifeguards
Youth summer camps	Low-cost, infrastructure-free safety net
Rural / off-grid recreational waters	No dependency on cellular or Wi-Fi
Flash-flood river corridors	“Flash Flood Alley” evacuation tracking
Search-and-rescue agencies	Real-time dispatch from live telemetry

Table 11 — Target deployment applications.

Thank You

Questions & Discussion

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